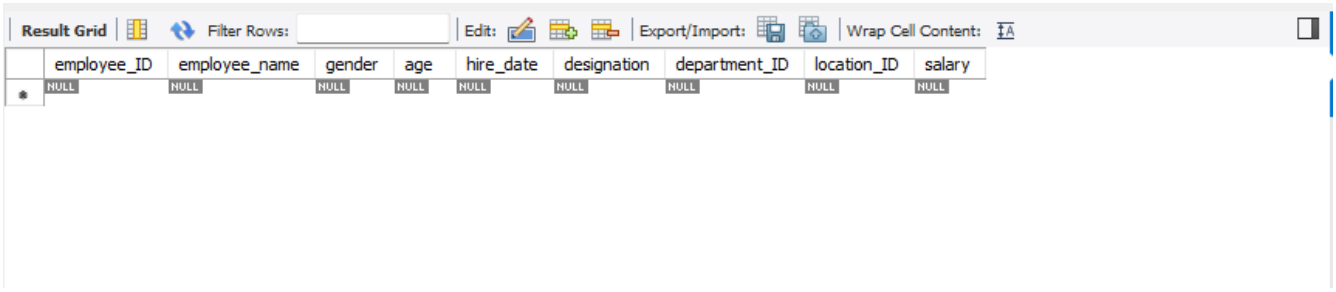
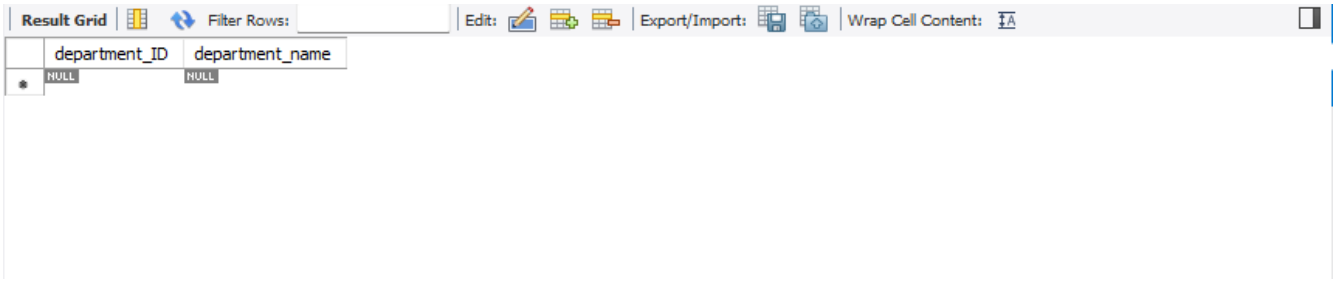
Assignment 6 – Clauses and Joins

Tables

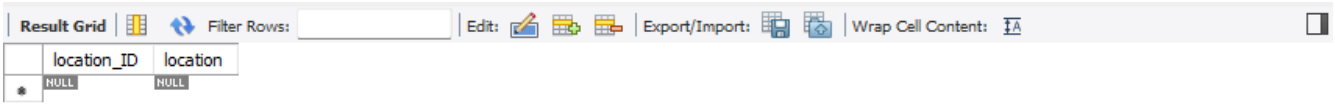
Employees:



Departments:

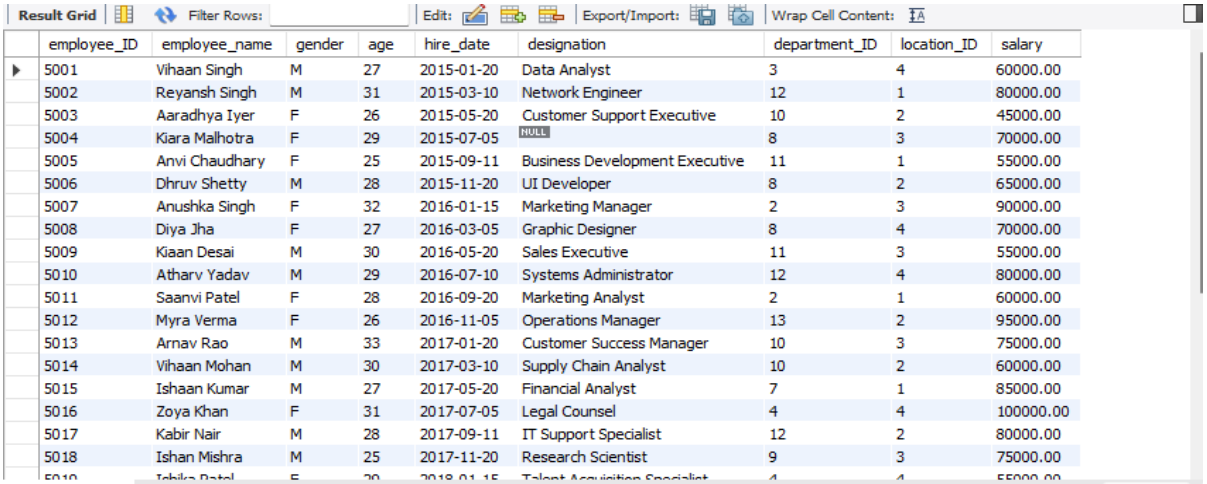


Locations:



- Used DML insert codes to populate the tables.

Employees:



employee_ID	employee_name	gender	age	hire_date	designation	department_ID	location_ID	salary
5001	Vihaan Singh	M	27	2015-01-20	Data Analyst	3	4	60000.00
5002	Reyansh Singh	M	31	2015-03-10	Network Engineer	12	1	80000.00
5003	Aaradhya Iyer	F	26	2015-05-20	Customer Support Executive	10	2	45000.00
5004	Kiara Malhotra	F	29	2015-07-05	NULL	8	3	70000.00
5005	Anvi Chaudhary	F	25	2015-09-11	Business Development Executive	11	1	55000.00
5006	Dhruv Shetty	M	28	2015-11-20	UI Developer	8	2	65000.00
5007	Anushka Singh	F	32	2016-01-15	Marketing Manager	2	3	90000.00
5008	Diya Jha	F	27	2016-03-05	Graphic Designer	8	4	70000.00
5009	Kiaan Desai	M	30	2016-05-20	Sales Executive	11	3	55000.00
5010	Atharv Yadav	M	29	2016-07-10	Systems Administrator	12	4	80000.00
5011	Saanvi Patel	F	28	2016-09-20	Marketing Analyst	2	1	60000.00
5012	Myra Verma	F	26	2016-11-05	Operations Manager	13	2	95000.00
5013	Arnav Rao	M	33	2017-01-20	Customer Success Manager	10	3	75000.00
5014	Vihaan Mohan	M	30	2017-03-10	Supply Chain Analyst	10	2	60000.00
5015	Ishaan Kumar	M	27	2017-05-20	Financial Analyst	7	1	85000.00
5016	Zoya Khan	F	31	2017-07-05	Legal Counsel	4	4	100000.00
5017	Kabir Nair	M	28	2017-09-11	IT Support Specialist	12	2	80000.00
5018	Ishan Mishra	M	25	2017-11-20	Research Scientist	9	3	75000.00

Departments:

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

department_ID	department_name
11	Business Development
6	Content Creation
10	Customer Support
3	Data Science
8	Design
7	Finance
4	Human Resources
12	IT
2	Marketing
13	Operations
5	Product Management
9	Research and Develo...
1	Software Development
NULL	NULL

Locations:

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

location_ID	location
2	Bangalore
1	Chennai
3	Hyderabad
4	Pune
NULL	NULL

1. Distinct Values:
   - Used DISTINCT to retrieve distinct salaries.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

salary
60000.00
80000.00
45000.00
70000.00
55000.00

2. Alias (AS):
   - Provided aliases for the "age" and "salary" columns as "Employee_Age" and "Employee_Salary", respectively.

```
65 •   SELECT age AS Employee_Age FROM employees;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Employee_Age
27
31
26
29
25

```
67 •  SELECT salary AS Employee_Salary FROM employees;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Employee_Salary
60000.00
80000.00
45000.00
70000.00
55000.00

3. Where Clause & Operators:

- Retrieve employees with a salary greater than ₹50000 and hired before 2016-01-01.

```
69 •  SELECT * FROM employees WHERE salary > 50000 AND hire_date < '2016-01-01';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_ID	employee_name	gender	age	hire_date	designation	department_ID	location_ID	salary
5001	Vihaan Singh	M	27	2015-01-20	Data Analyst	3	4	60000.00
5002	Reyansh Singh	M	31	2015-03-10	Network Engineer	12	1	80000.00
5004	Kiara Malhotra	F	29	2015-07-05	NULL	8	3	70000.00
5005	Anvi Chaudhary	F	25	2015-09-11	Business Development Executive	11	1	55000.00
5006	Dhruv Shetty	M	28	2015-11-20	UI Developer	8	2	65000.00

- Find the employee whose designation is missing and fill it with "Data Scientist".

Before:

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_ID	employee_name	gender	age	hire_date	designation	department_ID	location_ID	salary
5001	Vihaan Singh	M	27	2015-01-20	Data Analyst	3	4	60000.00
5002	Reyansh Singh	M	31	2015-03-10	Network Engineer	12	1	80000.00
5003	Aaradhya Iyer	F	26	2015-05-20	Customer Support Executive	10	2	45000.00
5004	Kiara Malhotra	F	29	2015-07-05	NULL	8	3	70000.00
5005	Anvi Chaudhary	F	25	2015-09-11	Business Development Executive	11	1	55000.00
5006	Dhruv Shetty	M	28	2015-11-20	UI Developer	8	2	65000.00
5007	Anushka Singh	F	32	2016-01-15	Marketing Manager	2	3	90000.00
5008	Diya Jha	F	27	2016-03-05	Graphic Designer	8	4	70000.00
5009	Kiaan Desai	M	30	2016-05-20	Sales Executive	11	3	55000.00
5010	Atharv Yadav	M	29	2016-07-10	Systems Administrator	12	4	80000.00
5011	Saanvi Patel	F	28	2016-09-20	Marketing Analyst	2	1	60000.00
5012	Myra Verma	F	26	2016-11-05	Operations Manager	13	2	95000.00
5013	Arnav Rao	M	33	2017-01-20	Customer Success Manager	10	3	75000.00
5014	Vihaan Mohan	M	30	2017-03-10	Supply Chain Analyst	10	2	60000.00
5015	Ishaan Kumar	M	27	2017-05-20	Financial Analyst	7	1	85000.00
5016	Zoya Khan	F	31	2017-07-05	Legal Counsel	4	4	100000.00
5017	Kabir Nair	M	28	2017-09-11	IT Support Specialist	12	2	80000.00
5018	Ishan Mishra	M	25	2017-11-20	Research Scientist	9	3	75000.00

After:

```
69 •  SELECT * FROM employees WHERE salary > 50000 AND hire_date < '2016-01-01';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_ID	employee_name	gender	age	hire_date	designation	department_ID	location_ID	salary
5001	Vihaan Singh	M	27	2015-01-20	Data Analyst	3	4	60000.00
5002	Reyansh Singh	M	31	2015-03-10	Network Engineer	12	1	80000.00
5004	Kiara Malhotra	F	29	2015-07-05	NULL	8	3	70000.00
5005	Anvi Chaudhary	F	25	2015-09-11	Business Development Executive	11	1	55000.00
5006	Dhruv Shetty	M	28	2015-11-20	UI Developer	8	2	65000.00

SORTING AND GROUPIN DATA

1. ORDER BY:

- Find employees sorted by department ID in ascending order and salary in descending order.

```
77 •  SELECT * FROM employees ORDER BY department_ID ASC;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_ID	employee_name	gender	age	hire_date	designation	department_ID	location_ID	salary
5020	Aarav Nair	M	32	2018-03-05	Software Engineer	1	1	90000.00
5007	Anushka Singh	F	32	2016-01-15	Marketing Manager	2	3	90000.00
5011	Saanvi Patel	F	28	2016-09-20	Marketing Analyst	2	1	60000.00
5023	Anika Paul	F	30	2018-09-20	Public Relations Specialist	2	2	70000.00
5001	Vihaan Singh	M	27	2015-01-20	Data Analyst	3	4	60000.00
5025	Avni Iyengar	F	31	2019-01-20	Data Scientist	3	4	100000.00
5026	Vivaan Singh	M	29	2019-03-10	Business Analyst	3	2	75000.00

```
79 •  SELECT * FROM employees ORDER BY salary DESC;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_ID	employee_name	gender	age	hire_date	designation	department_ID	location_ID	salary
5016	Zoya Khan	F	31	2017-07-05	Legal Counsel	4	4	100000.00
5025	Avni Iyengar	F	31	2019-01-20	Data Scientist	3	4	100000.00
5012	Myra Verma	F	26	2016-11-05	Operations Manager	13	2	95000.00
5024	Aryan Shetty	M	27	2018-11-05	Product Manager	5	1	95000.00
5007	Anushka Singh	F	32	2016-01-15	Marketing Manager	2	3	90000.00
5020	Aarav Nair	M	32	2018-03-05	Software Engineer	1	1	90000.00
5030	Sara Iyer	F	28	2019-11-20	Project Manager	5	1	90000.00

2. LIMIT:

- Display the first 5 employees hired in the year 2018.

```
81 •  SELECT * FROM employees WHERE YEAR(hire_date) = 2018 ORDER BY hire_date ASC LIMIT 5;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_ID	employee_name	gender	age	hire_date	designation	department_ID	location_ID	salary
5019	Ishika Patel	F	29	2018-01-15	Talent Acquisition Specialist	4	4	55000.00
5020	Aarav Nair	M	32	2018-03-05	Software Engineer	1	1	90000.00
5021	Advik Kapoor	M	26	2018-05-20	Finance Analyst	7	3	85000.00
5022	Aadhya Iyengar	F	28	2018-07-10	HR Specialist	4	4	60000.00
5023	Anika Paul	F	30	2018-09-20	Public Relations Specialist	2	2	70000.00
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

3. Aggregate Functions:

- Calculate the sum of all salaries in the Finance department.

```
83 •  SELECT SUM(salary) FROM employees Where department_ID = 7 ;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

SUM(salary)
170000.00

Department ID of finance department is 7.

- Find the minimum age among all employees.

```
85 •    SELECT MIN(age) FROM employees;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

MIN(age)
25

4. GROUP BY:

- List the maximum salary for each location.

```
190 •    SELECT location_ID,MAX(salary) FROM employees GROUP BY location_ID;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

location_ID	MAX(salary)
1	95000.00
2	95000.00
3	90000.00
4	100000.00

- Calculate the average salary for each designation containing the word 'Analyst'.

```
192 •    SELECT designation,AVG(salary) FROM employees WHERE designation LIKE '%Analyst' GROUP BY designation;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

designation	AVG(salary)
Data Analyst	60000.000000
Marketing Analyst	60000.000000
Supply Chain Analyst	60000.000000
Financial Analyst	85000.000000
Finance Analyst	85000.000000

5. HAVING:

- Find departments with less than 3 employees.

```
194 •    SELECT Department_ID, COUNT(*) FROM employees GROUP BY department_ID HAVING COUNT(*) < 3;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Department_ID	COUNT(*)
6	2
7	2
9	1
11	2
13	1

- Find locations with female employees whose average age is below 30.

```
196 •  SELECT Location_ID, AVG(age) FROM employees GROUP BY location_ID HAVING AVG(age) < 30;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Location_ID	AVG(age)
1	28.0000
2	28.7500
3	29.5714
4	28.8571

JOINS

1. INNER JOIN:

- List employee names, their designations, and department names where employees are assigned to a department.

```
198 •  SELECT e.employee_name,e.designation,d.department_name
199    FROM employees e
200    INNER JOIN departments d
201    ON e.department_ID = d.department_ID;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

employee_name	designation	department_name
Anvi Chaudhary	Business Development Executive	Business Development
Kiaan Desai	Sales Executive	Business Development
Ananya Paul	Content Writer	Content Creation
Anaya Kapoor	Event Coordinator	Content Creation
Aaradhya Iyer	Customer Support Executive	Customer Support

Result 12

2.LEFT JOIN:

- List all departments along with the total number of employees in each department, including departments with no employees.

```
204 •  SELECT d.department_ID,d.department_name, count(e.department_ID)
205    AS total_number_of_employees
206    FROM departments d
207    LEFT JOIN employees e
208    ON d.department_ID = e.department_ID GROUP BY d.department_ID;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

department_ID	department_name	total_number_of_employees
9	Research and Development	1
10	Customer Support	3
11	Business Development	2
12	IT	4
13	Operations	1

Result 17

3.RIGHT JOIN:

- Display all locations along with the names of employees assigned to each location. If no employees are assigned to a location, display NULL for employee name.

```
210 •   SELECT l.location_ID, l.location, e.employee_name
211     FROM locations l
212     RIGHT JOIN employees e
213     ON l.location_ID = e.location_ID ORDER BY l.location_ID ASC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

location_ID	location	employee_name
1	Chennai	Reyansh Singh
1	Chennai	Anvi Chaudhary
1	Chennai	Saanvi Patel
1	Chennai	Ishaan Kumar
1	Chennai	Aarav Nair

Result 19